Exam 2 – Practice 3 Math 110.

Exam Guidelines This is an in-class, written exam with a 75-minute time limit.

- **Permitted Materials:** You may use a basic calculator and formula sheets.
- **Formula Sheet Restrictions:** Your sheets must contain formulas only; no examples or worked problems are permitted. All sheets will be inspected at the start of the exam.
- **Prohibited Items:** Phones and all other smart devices are strictly forbidden.
- **Academic Integrity:** The use of AI is prohibited. Any AI usage will result in an automatic F for the exam and may lead to failing the entire course.
- Show **ALL** your work for credits.

1. Solve each quadratic by factoring or the quadratic formula.
   a. $x^2 = 4$

      $x = 2;\ x = -2$

   b. $2x^2 - 12x + 18 = 0$

      $x = 3$

   c. $-3x^2 - 6x + 9 = 0$

      $x = -3;\ x = 1$

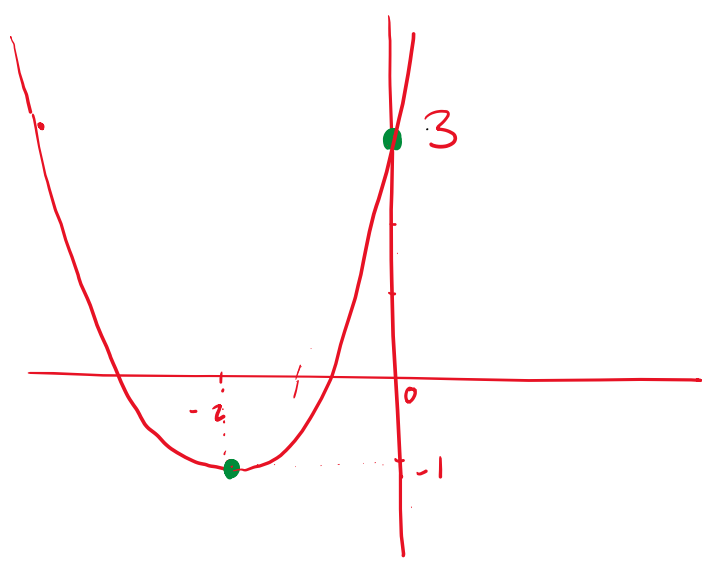
   d. $x^2 + 3x = 2x + 6$

      $x = -3;\ x = 2$

2. Graph of the quadratic functions. Label the vertex and another point.
   a. $y = x^2 + 4x + 3$

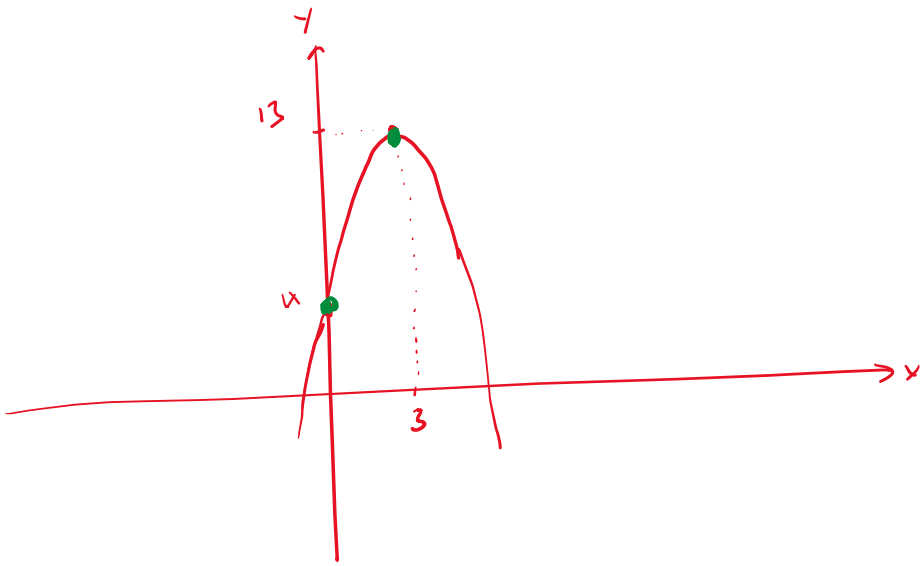
      Vertex $(-2; -1)$

      Another point $(0, 3)$



b. $y = -x^2 + 6x + 4$

Vertex $(3; 13)$ Another point $(0, 4)$



3. Suppose that in a monopoly market the total cost per week of producing a high-tech product is given by $C = 1800 + 80q + q^2$. Suppose further that the sales price function for this product is
$p = 260 - 2q$.

a. Find the revenue function in term of q.

$R = q \cdot (260 - 2q) = 260q - 2q^2$

b. Find the number of units that will **maximize the revenue.**

$q = 65$

c. Find the profit function

Profit $= R - C = -3q^2 + 180q - 1800$

d. Find the number of units that will give **break-even** for the product

$$-3q^2 + 180q - 1800 = 0$$

$$q = \frac{-180 \pm \sqrt{180^2 - 4 \cdot (-3) \cdot (-1800}}{-3 \cdot 2}$$

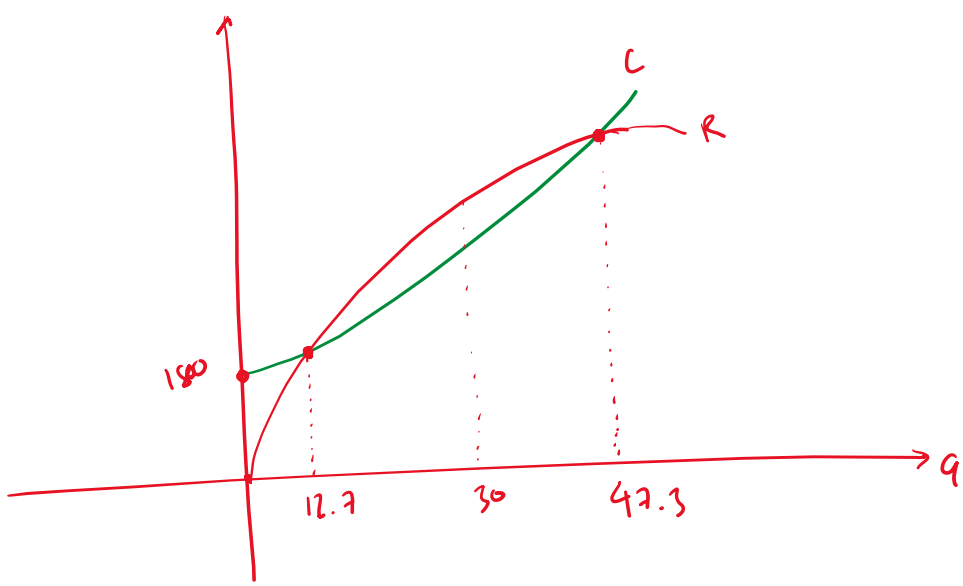
$$q \approx 12.7 \; ; \quad q = 47.3$$

e. Find **the maximum profit** and the number of products need to maximize the profit.

$$q = 30$$

Max profit $= 900$

f. Graph the revenue function and the cost function label the break-even points, fixed cost, and the maximized profit point.



4. On a certain route, an airline carries 5000 passengers per month, each paying \$30. A market survey indicates that for each \$1 increase in the ticket price, the airline will lose 80 passengers.

   a. What is the airline's current revenue?

   $R = 150\,000$

   b. Create an income (revenue) function if "x" is defined as the number of \$1 price increases

   $R = (30 + x)(5000 - 80x)$

   c. Find the number of \$1 price increases that will maximize the revenue.

   $x = 16.25$

   d. Find the new ticket price (that will maximize the revenue)

   new price $= 30 + x = 46.25$

   e. Find the number of passengers at that price in d.

   $5000 - 80 * x = 3700$

   f. Find the new maximum income (income at that price in d)

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5. If the supply function for a commodity is given by $p = 5q^2 + 10q$ and the demand function is given by $p = 200 - 5q^2$, find the point of market equilibrium (Supply equals Demands).

$$5q^2 + 10q = 200 - 5q^2$$

$$\Leftrightarrow \quad 10q^2 + 10q - 200 = 0$$

$$\Leftrightarrow \quad q^2 + q - 20 = 0$$

$$q = 4$$

$$p = 200 - 5q^2 = 120$$